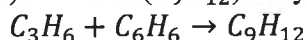
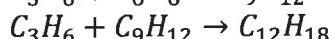
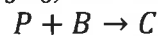


To receive full credit, you must show all of your work.

- (1) (60 pts.) Cumene (C_9H_{12}) is synthesized from propylene (C_3H_6) and benzene (C_6H_6):



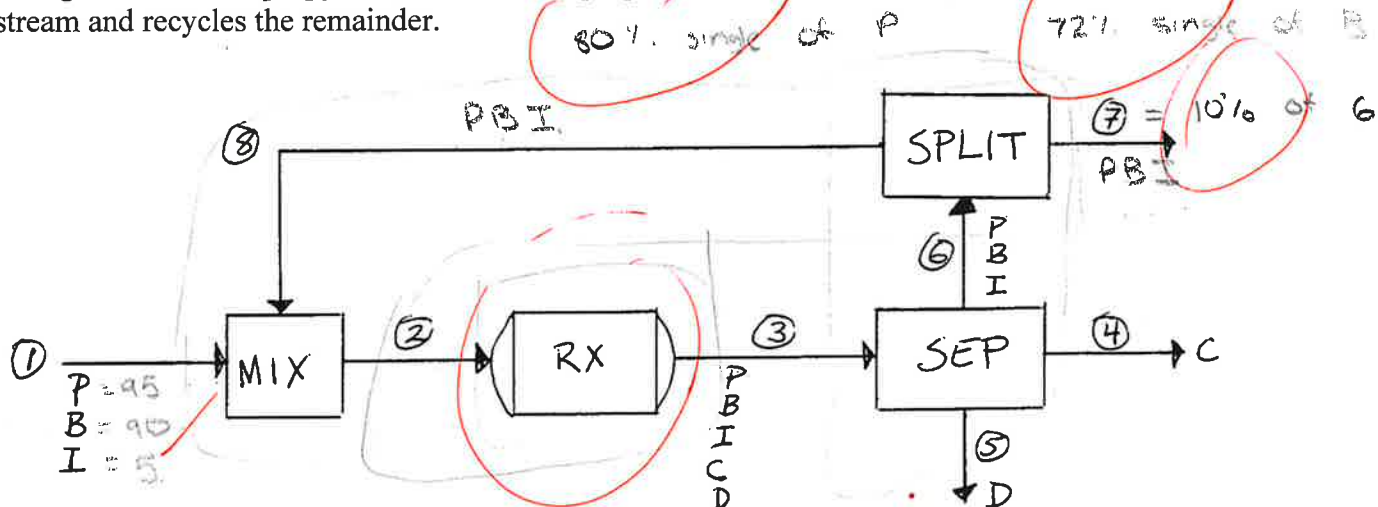
(R1)



(R2)



The undesired byproduct diisopropylbenzene ($C_{12}H_{18}$) is produced in a second competing reaction. Refer to the block flow diagram below. The fresh feed contains 95 mol/s propylene, 90 mol/s benzene, and 5 mol/s of propane I (C_3H_8) which is inert. This stream is combined in the mixer with the recycle stream. The reactor single-pass conversion is 80% of the propylene and 72% of the benzene. The reactor product goes to a separator which has three output streams: a stream of pure cumene, a stream of pure diisopropylbenzene, and a stream containing the unreacted propylene, benzene, and propane, which is sent to a splitter. The splitter purges 10% of this stream and recycles the remainder.



- (a) Finish labeling the flow diagram above. In the column "Entire Process" (see Table below), use the degree of freedom analysis (see p. 105 of Murphy) to determine if the problem is well posed. Give a brief explanation for each entry. (10 pts.)

	Entire Process	Explanation
4a # stream variables	22 ✓	# of components in 1, 2, 3, 4, 5, 6, 7, 8 (3)(3)(5)(10)(3)(3)(3) = 22
4b # reaction variables	2 ✓	2 rxns, one wanted one undesired
# indep variables	24 ✓	total
5a # specified flows	1 ✓	only stream 1
5b # spec stream comps and split	10 + 1 = 11 ✓	2 outputs in splitter → add 1
5c # performance specs	3 ✓	split = 10%, single p conversion given for P and also B
5d # material balances	4 ✓	mats are P, B, C, and D
# indep eqns	19 ✓	total
DoF	5 ✓	22 - 19

Problem 2 continued

- (b) Calculate the fractional yield for cumene for the following percent excesses in benzene: 0%, 100%, and 400%. Be careful to avoid roundoff error. (15 pts.)

$$O = \frac{B_{e,} - B_i}{B_i}$$

$$\text{yield} = \frac{\text{formed}}{\text{fed}} =$$

at 0% B_i

2

% Excess Benzene	0%	100%	400%
Fractional Yield			

- (c) Suggest how the design of the process could be changed to improve the fractional yield of cumene. (5 pts.)

heat up process, add more P and D
in reactor, create larger recycle stream

24

74

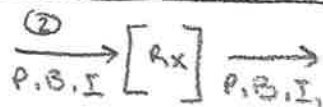
$$P_3 = 0.8(0.9P_3 + P_1) = 0.18P_3 + 0.2P_1$$

$$P_3 = \frac{0.2P_1}{0.82} = 23.17$$

Problem 1 continued

- (b) Write the appropriate balance equations. Calculate the molar flow rates in the process in mol/s. Be sure to define the system for any sets of balance equations you write! Fill in the Table below with your results. (45 pts.)

Reactor balance ^{2 mixer}



$$P_3 = P_2 - \dot{E}_1 - \dot{E}_2$$

$$B_3 = B_2 - \dot{E}_1$$

$$C_3 = \dot{E}_1 - \dot{E}_2$$

$$D_3 = \dot{E}_2$$

$$I_3 = I_2$$

$$B_3 = .72(B_2)$$

$$B_3 = .72(.9B_3 + B_1)$$

$$B_3 = .648B_3 + 64.8$$

$$B_3 = 184.09$$

$$B_2 = .9(184.09 + 90)$$

$$184.09 = 246.681 - \dot{E}_1$$

$$\dot{E}_1 = 62.591 = 71.591$$

$$271.429 = 334.286 - 62.591 - \dot{E}_2$$

$$\dot{E}_2 = 0.266 = -3.7338$$

$$D_3 = 0.266$$

$$C_3 = 62.591 - 0.266$$

Stream Summary Table: All flows in mol/s

J/Stream →	1	2	3	4	5	6	7	8
P	95	334.286	271.429	-----	-----	271.429	27.143	244.287
B	90	246.681	184.09	-----	-----	184.09	18.401	165.689
I	5	50	50	-----	-----	50	5	45
C	-----	-----	62.325	62.325	-----	-----	-----	-----
D	-----	-----	0.266	-----	0.266	-----	-----	-----

- (c) Calculate the overall fractional conversions of propylene, the fractional selectivity, and the fractional yield

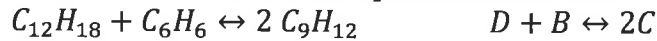
(5 pts.) overall conv = $\frac{95 - 27.143}{95} = .714$

yield = $\frac{27.143}{95} = .28$

selectivity = $\frac{62.325}{(.266 + 5 + 18.401 + 27.143)} = 1.36 > 1?$

(42)

- (2) (40 pts.) In the production of cumene, diisopropylbenzene is produced as a byproduct. To avoid wasting this material it can be converted to cumene in an equilibrium reaction:



A stream of 10 mol/s diisopropylbenzene is mixed with excess benzene and fed to an equilibrium reactor. The relation between the equilibrium constant and the mole fractions in the product stream is:

$$D = 10$$

$$K = \frac{x_C^2}{x_D x_B}$$

where under the reactor conditions $K = 0.0005$, which is a small value favoring the reactants.

- (a) Write the appropriate species balance equations for this reaction. Combine these results with the equilibrium constant relation to derive a general expression for the percent excess benzene. (20 pts.)

$$D_e = D_i - \xi \quad B_e = B_i - \xi \quad C_e = 0 + 2\xi$$

$$D_e = 10 - \xi$$

$$K = \frac{x_C^2}{x_D x_B}$$

$$0.0005 = \frac{(2\xi)^2}{(10-\xi)(B_i-\xi)} \cdot \frac{1}{(10+B_i)^2}$$

$$\text{total} = 10 - \xi + 2\xi + B_i - \xi = 10 + B_i$$

$$10 + B_i = \text{total}$$

$$\frac{B_i - B_n}{B_i} = \% \text{ excess} = \frac{4\xi}{10+B_i} = \frac{(2\xi)^2}{(10-\xi)(B_i-\xi)} = \frac{1}{2000}$$

$$4\xi^2 = 1 \quad \xi = .5$$

$$(10 - .5)(B_i - .5) = 2000$$

$$10B_i - 5 - .5B_i + .25 = 2000$$

$$9.5B_i = 2004.75$$

$$B_i = 211.0263$$

$$B_e = 211.0263 - \xi = 210.526$$

$$\% \text{ excess} = \frac{210.526}{211.0263} = .9976$$

$$\boxed{99.76\%}$$